TEXORA · SESSIONAL NOTES · PRACTICAL STUDY SHEET

Practical Study of the Machinery Involved: Circular Weft, Flat and Warp Knitting Machines

A working guide to the three knitting machine families in the laboratory: what to identify at each machine, which settings to record, and the production figures your report must carry — with the circular machine and its needle–sinker–cam arrangement drawn in detail.

TE-3102 · KNITTING TECHNOLOGY SESSIONAL · STUDY 1 · LEVEL 3 TERM 1

1. Object and method of the study

Object: to connect knitting theory with the machines themselves — to identify every working part of the circular weft, flat (V-bed) and warp knitting machines, read their settings as the laboratory has set them, and record the production figures each setting produces.

Method — the same routine at every machine: (1) **trace the material path** from yarn package to fabric roll, naming every element touched; (2) **identify the working parts** listed below, stating in one line what each does; (3) **read and record the settings** — plate data (diameter, gauge, feeders or working width), speeds, stitch-length setting — observed, never altered, and only with the instructor’s permission; (4) **record production points** (Section 2.2 worked example); (5) **sketch** the path (Figures 1–2 as models) and note faults seen in operation, with likely causes.

2. The circular weft knitting machine

The laboratory circular machine knits weft fabric in tube form: needles stand vertically in the cylinder, one feeder follows another around the circle, and every feeder lays one course. Single jersey machines carry needles in the cylinder only; double jersey (rib/interlock) machines add a second set in a horizontal dial above it.

2.1 Parts to identify

- **Creel and yarn feeders** — the creel holds the cones, a guide eye over each package; one storage/positive feed unit per feeder on a ring above the cylinder stores a reserve of yarn and delivers it at a controlled rate (Study 3).

- **Needle cylinder and latch needles** — steel cylinder with vertical slots (tricks), one latch needle per trick; on a spare needle identify hook, spring-loaded latch, stem, butt and tail — the butt is what the cams drive.
- **Sinkers, sinker ring and cylinder cams** — sinkers are the thin horizontal blades between the needles at the cylinder top, holding fabric down at clearing and supporting loops at knock-over. The cam box carries the track at each feeder: clearing cams lift the butts, the adjustable **stitch cam** draws needles down to set loop length, guard cams keep butts in the track; find the knit/tuck/miss positions where fitted.
- **Yarn carrier (feeder)** — the guide presenting yarn to the open hooks at the correct height and angle. **Dial and dial needles (double jersey only)** — horizontal disc above the cylinder carrying the second needle set radially, at right angles to the cylinder needles.
- **Take-down, batching, drive and stop motions** — rollers below the cylinder draw the tube down at a controlled rate to the batching roll at the base (Study 3); motor, drive and control panel; yarn-break, needle-break and press-off stop motions with signal lamps (Study 3).

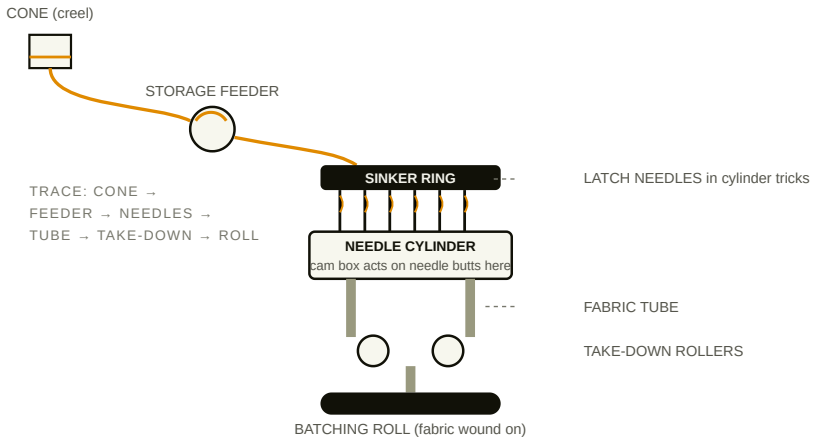


Figure 1 — General arrangement of a circular weft knitting machine (one feeder shown): cone → storage feeder → needle hooks at sinker-ring level → fabric tube drawn down inside the cylinder to take-down rollers and batching roll.

2.2 Settings and observations to record

ITEM	WHAT TO WRITE IN THE REPORT
Maker, model, machine type	From the maker’s plate; single or double jersey; direction of rotation
Cylinder diameter and gauge	Diameter in inches; gauge = needles per inch of circumference; compute the total needle number (worked example below)
Feeders; speed; yarn	Yarn carriers counted (fitted vs. in use); cylinder rpm from stopwatch or display; fibre, count (Ne or tex) and ends per feeder from the cone ticket

Stitch cam; take-down	Stitch-cam dial reading at one feeder (this setting fixes the stitch length); take-down roller type and drive, fabric batched on a roll or plaited
Fabric produced	Structure knitted (plain, rib, interlock…); tube width and course density by counting glass

WORKED EXAMPLE — PRODUCTION OF THE LABORATORY SINGLE JERSEY MACHINE (VERIFY EACH LINE)

Observed: diameter **30 in**, gauge **24**, **96 feeders**, speed **25 rpm**, yarn **30 Ne** cotton, stitch length **3.0 mm**.

Needles in the cylinder: N = π × D × G = 3.1416 × 30 × 24 = **2,262 needles** (approx.).

Yarn in one course: every needle draws one stitch length of yarn, so course length = 2,262 × 3.0 mm = 6,786 mm ≈ **6.79 m**.

Courses per minute: each revolution knits one course per feeder, so 96 × 25 = **2,400 courses/min**.

Yarn consumption: 6.79 m × 2,400 = **16,286 m/min** = 977,160 m/h. One pound of 30 Ne yarn measures 30 × 840 yd = 25,200 yd = 23,043 m, so consumption = 977,160 ÷ 23,043 ≈ **42.4 lb/h ≈ 19.2 kg/h**. Weigh a timed roll and compare: agreement within a few per cent confirms the stitch length and feeder count recorded.

2.3 The needle–sinker–cam arrangement in detail

Study one knitting point closely (machine stopped, inched by hand, with permission). The butt runs in the cam track: the clearing cam raises the needle so the old loop opens the latch and slips below it; yarn is fed into the open hook; the stitch cam draws the needle down — the old loop closes the latch, is knocked over the hook, and the new yarn is pulled through as a new loop. The sinker works in counter-rhythm: forward during clearing, back at knock-over (Figure 2; the full cycle is Study 2).

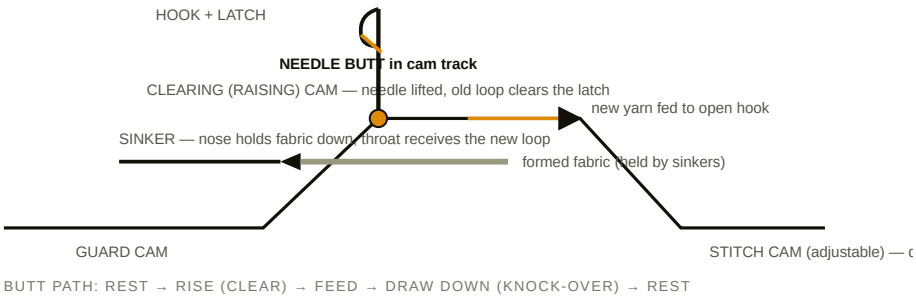


Figure 2 — Needle, sinker and cam at one knitting point: the cam track (guard, clearing and stitch cams) drives the butt through one loop-forming cycle while the sinker holds, then releases, the fabric.

3. The flat knitting machine (V-bed)

3.1 Parts to identify

- **Needle beds** — two flat beds set like an inverted V (about 90° between them); the needles face each other across the top gap, so rib and links-links (purl) structures can be knitted and loops transferred bed to bed.
- **Latch needles** — one set per bed, butts projecting into the carriage cam path; handle a spare and note the type actually fitted.
- **Carriage** — traverses the full bed width carrying the cam system; each traverse knits one course on the needles it selects. Identify stitch cams (loop length), raising cams, and the tuck/miss selection.
- **Yarn carriers** — run on a rail above the beds and lay yarn across the hooks ahead of the carriage; one carrier per yarn.
- **Brushes, holding-down elements, take-down** — brushes open the latches before feeding; holding-down sinkers or a take-down comb control the fabric at the start of a piece; rollers or a weighted comb below the beds draw the panel down.
- **Racking** — shifts one bed sideways by one or more needle spaces, for rib setting and pattern effects.

3.2 Settings and observations to record

ITEM	WHAT TO WRITE IN THE REPORT
Gauge and working width	Needles per inch per bed; bed width; total needles per bed = width × gauge
Structure; stitch cams	1×1 rib, full rib, plain on one bed, or shaped panel; beds and carriers working; stitch-cam dial settings for both beds, noting any difference and why
Carriage speed; take-down	Traverses per minute (one traverse = one course per system); take-down setting and how a new piece is started (comb and weights)

4. The warp knitting machine

Warp knitting differs in principle: yarns come from **warp beams** — one sheet of parallel ends per guide bar — and **each needle receives its own yarn from its own guide**, the guides lapping sideways to bind the wales together (Study 2). Record whether the laboratory machine is a tricot machine (two or more guide bars, fine gauge) or a Raschel machine (coarser, often with a fall plate).

4.1 Parts to identify

- **Warp beams** — large flanged beams at the back, one per guide bar, each wound with hundreds of parallel ends; find the let-off drive that releases warp at the set run-in.
- **Guide bars** — full-width bars each carrying a row of **guides** (steel blades with an eye at the tip), one per needle space; the bars swing through the needles and shog (shift sideways) to make the laps.
- **Needles, sinker bar and trick plate** — a complete needle bar of spring-beard, latch or compound needles moving as one unit, unlike individually cammed weft needles (on a compound machine, find the separate closing-wire bar); sinkers hold the loops down between the needles, and the trick plate supports the fabric at knock-over.
- **Pattern mechanism; take-up and batching** — pattern chain (links of different heights driving the shog) or electronic shog control — find where the lapping pattern is stored; take-up rollers draw fabric from the needles at a set rate to the batching roll at the front.

4.2 Settings and observations to record

ITEM	WHAT TO WRITE IN THE REPORT
Machine type and gauge	Tricot or Raschel; gauge as needles per inch (older tricot plates may state it per 2 inches — copy the plate exactly); bars fitted and threaded, ends per beam, which bar is front
Machine speed	Courses per minute; note one rack = 480 courses, the unit of warp-knitted production accounting
Run-in; lapping in use	Warp length fed per rack for each bar, as set on the let-off (different run-ins per bar are normal); the structure being knitted (e.g. two-bar tricot) and, if visible, the chain link arrangement

5. The three families compared

POINT	CIRCULAR WEFT	FLAT (V-BED)	WARP
Yarn supply	Cones; one yarn around all needles	Cones; one yarn across a bed	Warp beams; an end per needle per guide
Needle action	Driven individually by cams, one system per feeder	Driven by carriage cams, one course per traverse	All needles move together as one bar; guides lap the yarn
Typical fabric	Tubular jersey, rib, interlock	Flat panels, fully-fashioned garments	Open-width tricot, nets, laces

6. What the lab report must contain

For each machine: **identity** (plate data); **material path** traced, with a labelled sketch (Figures 1–2 as models); **parts table** (part, location, function in one line); **settings record** — Sections 2.2, 3.2 and 4.2 filled with *observed* values, never copied from the theory sheet; for the circular machine, the **production check** repeated from your own figures; and **faults observed**, with cause and remedy.

7. Practice (viva preparation)

Answer in your own words; brief model answers follow.

1. A circular machine has a 30-inch cylinder of gauge 24. How many needles does it carry, and why can the exact number differ slightly from your calculated figure?
2. Using the worked example of this sheet, how much yarn does the machine consume in one minute, and what single setting change would increase that figure without changing the speed?
3. Give two structural things a V-bed flat machine can do that a single-cylinder circular machine cannot, and name the machine parts that make each possible.
4. On the warp knitting machine, why is there one warp beam per guide bar, and what does the run-in setting control?

Answer key

1. N = π × 30 × 24 ≈ **2,262 needles**. The fitted number can differ by a few: the circumference must divide into whole tricks, and some positions may carry no needle, so the maker fits the nearest practical whole number.
2. Consumption = course length × courses per minute = 6.79 m × 2,400 = **16,286 m/min**. Increasing the **stitch length** (stitch-cam setting) puts more yarn in every course at unchanged speed; more feeders or higher rpm raise it too, but via the course rate, not the yarn per course.
3. (i) **Rib and purl structures** — the opposed beds, needles crossing at the top of the V, let loops be drawn to both sides. (ii) **Transfer and shaping** — opposed beds plus racking let loops move between needles and needles enter or leave work to widen or narrow the panel. (Accept also: a panel knitted to exact width, not a tube.)
4. Each bar laps independently and is consumed at its own rate, so each needs its own beam, let off at its own rate. The **run-in** — warp fed per rack (480 courses) — fixes that bar’s consumption and its share of fabric weight and tightness.